ViGoEmotions: Task and Dataset

Multi-label emotion recognition in Vietnamese comments.

20,664 comments; 27 emotions plus a Neutral state.

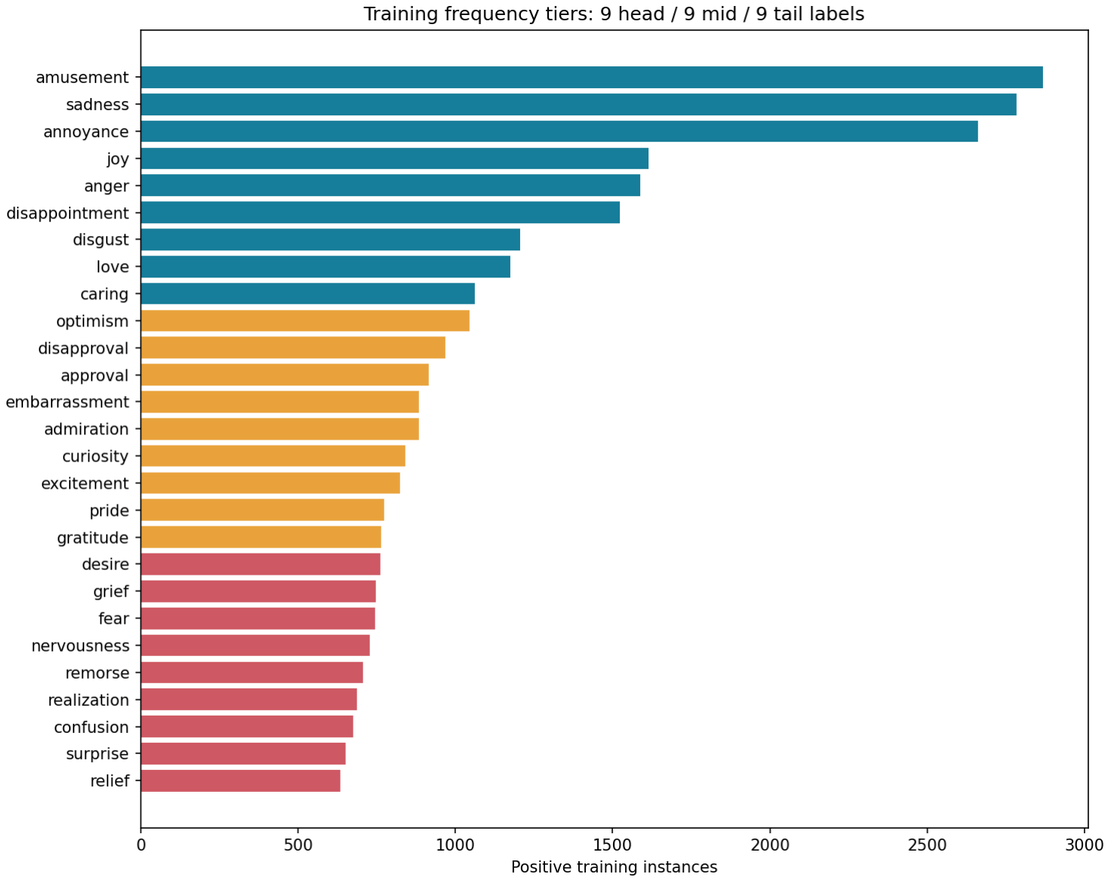
Effective train / validation / test: 16,530 / 2,066 / 2,067.

Three models; compare 27 and 28 outputs.

All model scores are validation results.

Data Profile and Leakage Prevention

- Mean 1.859 emotions; median 11 tokens / 47 characters.
- Amusement–relief frequency ratio: about 4.52×.
- Fixed split; one overlapping training sample removed.
- Fit vocabulary/IDF on train only; no model test scores.



27 vs 28 Outputs: What Changes?

27 outputs: Neutral-only has an all-zero emotion target.

28 outputs: Neutral becomes a separate learned label.

Keep mixed annotations; no Neutral exclusion rule.

Validation: 75 Neutral-only + 19 mixed = 94.

Independent sigmoid outputs with binary cross-entropy.

Models and Shared Training Protocol

TF-IDF + OvR LR: liblinear; max_iter = 1,000.

BiGRU from scratch: hidden 128; batch 64; lr 0.001.

PhoBERT full fine-tuning: batch 16; lr 2e-5.

Both neural models: 20 epochs per label setting; seed 42.

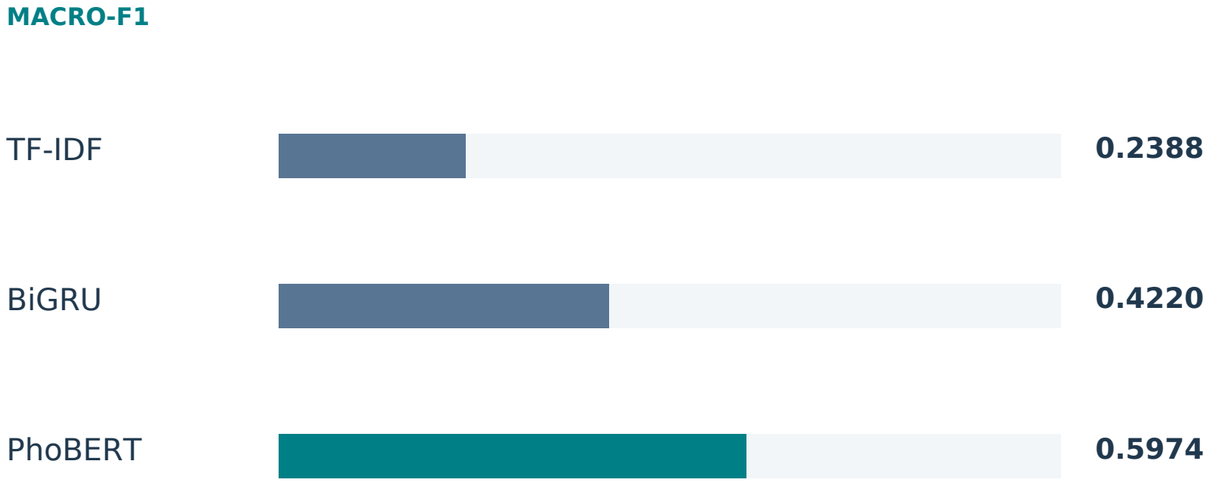
Select checkpoints by Macro-F1 on the same 27 emotions.

Results: 27 Emotions

TF-IDF Macro-F1: 0.2388.

BiGRU Macro-F1: 0.4220.

PhoBERT Macro-F1: 0.5974; Micro-F1: 0.5996.



PhoBERT Hamming Loss: 0.0550; LRAP: 0.7419.

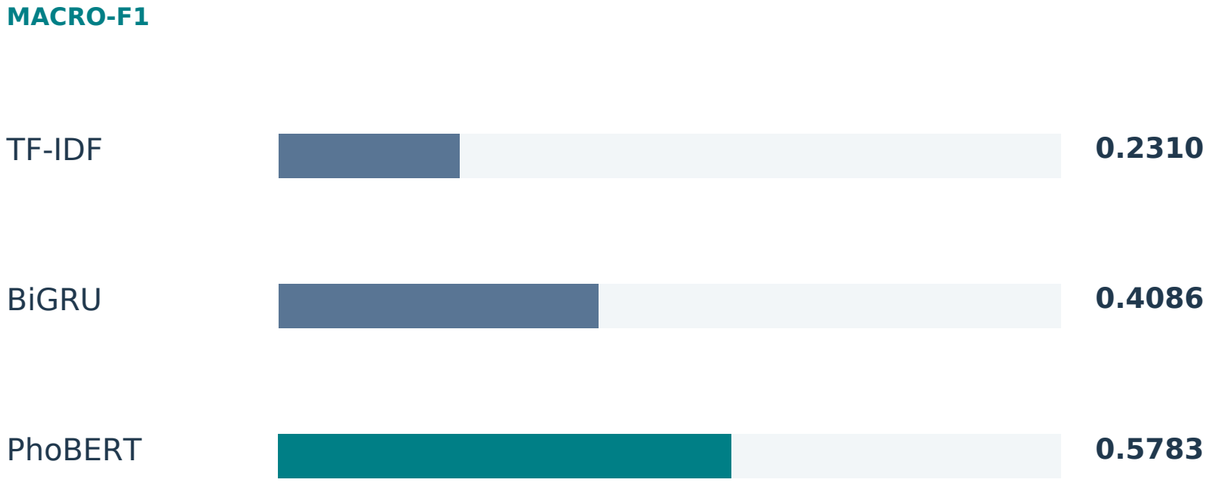
Higher F1/LRAP is better; lower Hamming Loss is better.

Results: 28 Labels Including Neutral

TF-IDF Macro-F1: 0.2310.

BiGRU Macro-F1: 0.4086.

PhoBERT Macro-F1: 0.5783; Micro-F1: 0.5829.



PhoBERT Hamming Loss: 0.0575; LRAP: 0.7077.

These metrics evaluate all 28 labels.

Does Neutral Help the Shared 27 Emotions?

PhoBERT with 27 outputs: 0.5974.

PhoBERT with 28 outputs, scored on 27: 0.5836.

Difference: -1.38 percentage points.

No observed improvement on the shared emotions.

One seed; no causal or statistical conclusion.

Neutral Detection Remains Difficult

PhoBERT Neutral F1: 0.4369.

Precision: 0.4018; recall: 0.4787.

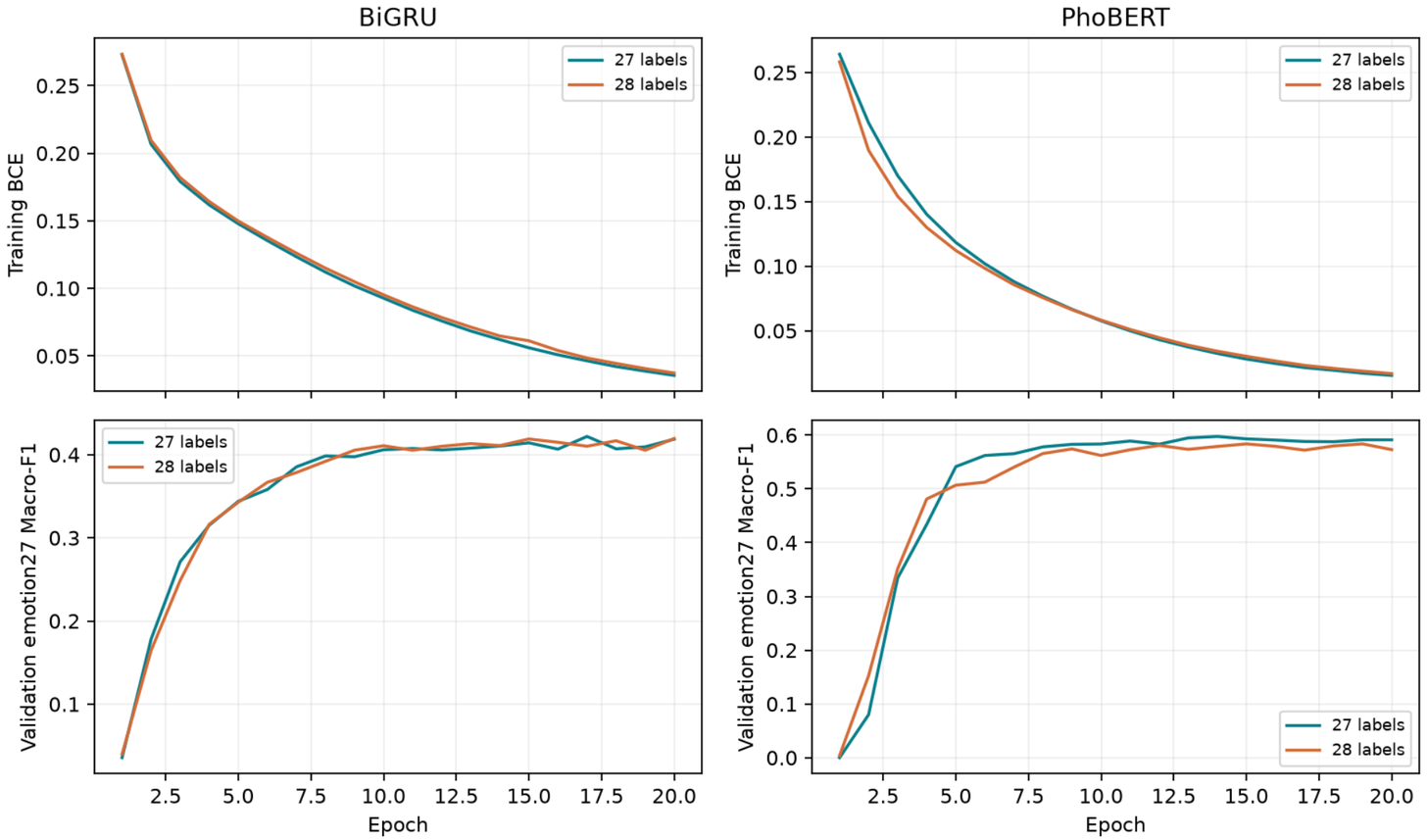
45 TP / 67 FP / 49 FN; support: 94.

The 27-output model has no Neutral classifier.

An empty prediction is not automatically Neutral.

Learning Curves and Best Checkpoints

- All four neural runs completed epochs 1–20.
- Best BiGRU epochs, 27/28: 17 / 20.
- Best PhoBERT epochs, 27/28: 14 / 15.
- Select the best checkpoint, not necessarily the last.
- Training loss alone cannot confirm overfitting.



Q1–Q2: Per-Label Performance and Frequency

Q1

PhoBERT best label: gratitude, F1 0.8370.

Lowest label: disapproval, F1 0.3623.

Q2

Training frequency vs F1: Spearman rho 0.018.

p-value 0.928; descriptive evidence only.

Complete per-label results are in the report.

Q3–Q4: Label Pairs and Text Styles

Q3

Top training pair: disappointment–sadness (904 cases).

Both correct: 63/122 validation cases; joint recall 51.64%.

Q4

Largest PhoBERT–BiGRU style gap: teencode.

Macro-F1 gap: +22.13 percentage points.

Style groups are heuristic and may overlap.

Q5: Does the Default Threshold Matter?

PhoBERT 27 at 0.5: Macro-F1 0.5974.

Best validation grid point: 0.35, F1 0.5978.

Same checkpoint; exploratory validation analysis.

Very small F1 gain; Hamming Loss increases.

Main results remain at threshold 0.5.

Error Analysis: New Neutral Predictions

Six cases: three Neutral false positives and three misses.

Potential issues: implied emotion, short text, HTML, profanity.

IDs and abstract observations; no raw comments.

Preliminary AI-assisted review; team confirmation pending.

Final Research Question and Next Steps

Is one threshold suitable for every emotion?

Freeze the best neural checkpoint and its weights. → Compare 0.5, global tuned and per-label thresholds. → Tune on validation; lock choices; evaluate test once.

Review at least 20 false negatives and 20 false positives.

Conclusions and Evidence

PhoBERT leads within both label settings.

Neutral does not improve shared-emotion F1 in this run.

Evidence: Kaggle 08/09, saved predictions and manifests.

Limits: one seed, validation selection, no test scores.

Sources: course brief, ViGoEmotions, PhoBERT and libraries.

Supplementary Table A · Model performance

Validation n = 2,066 · threshold 0.5 · full-label scores reported separately

Labels	Model	Macro-F1 ↑	Micro-F1 ↑	Hamming ↓	LRAP ↑	Best ep.	Train time
27	TF-IDF + OvR LR	0.2388	0.2712	0.0606	0.6407	—	2.75 s
27	BiGRU	0.4220	0.4590	0.0626	0.6191	17	80.3 s
27	**PhoBERT**	**0.5974**	**0.5996**	**0.0550**	**0.7419**	14	4.05 h
28	TF-IDF + OvR LR	0.2310	0.2662	0.0601	0.6096	—	2.52 s
28	BiGRU	0.4086	0.4483	0.0628	0.5851	20	83.6 s
28	**PhoBERT**	**0.5783**	**0.5829**	**0.0575**	**0.7077**	15	3.90 h

Both neural models trained for 20 epochs. Best ep. is the selected checkpoint. Rank models within each label block; full 27-label and 28-label Macro-F1 use different label sets.

Source: current model_comparison.csv. CPU TF-IDF; Tesla T4 neural. Times exclude preprocessing; no test scores.

Supplementary Table B · Pair and text-style analysis

27-output study only · validation · threshold 0.5 · supplementary detail for slide 11

DISAPPOINTMENT + SADNESS · TRAINING COUNT 904

Model	Joint TP	Support	Recall
TF-IDF + OvR LR	2	122	1.64%
BiGRU	37	122	30.33%
PhoBERT	**63**	122	**51.64%**

TEXT-STYLE MACRO-F1

Group	n	BiGRU F1	PhoBERT F1	Δ (pp)
All	2,066	0.4220	0.5974	+17.54
Emoji-heavy	199	0.3506	0.4789	+12.83
Short	343	0.4303	0.5654	+13.51
Teencode	307	0.3333	0.5547	**+22.13**

Joint recall counts both emotions predicted among examples annotated with both; extra predicted labels are allowed. Style groups overlap and differ in size; Δ = PhoBERT − BiGRU in Macro-F1 percentage points. One seed; no causal or significance claim.

Sources: current pair_preservation.csv; neural_style_comparison.csv. Pair selected by training co-occurrence; no test metrics.